Advanced spatial data visualization

CS424: Visualization & Visual Analytics

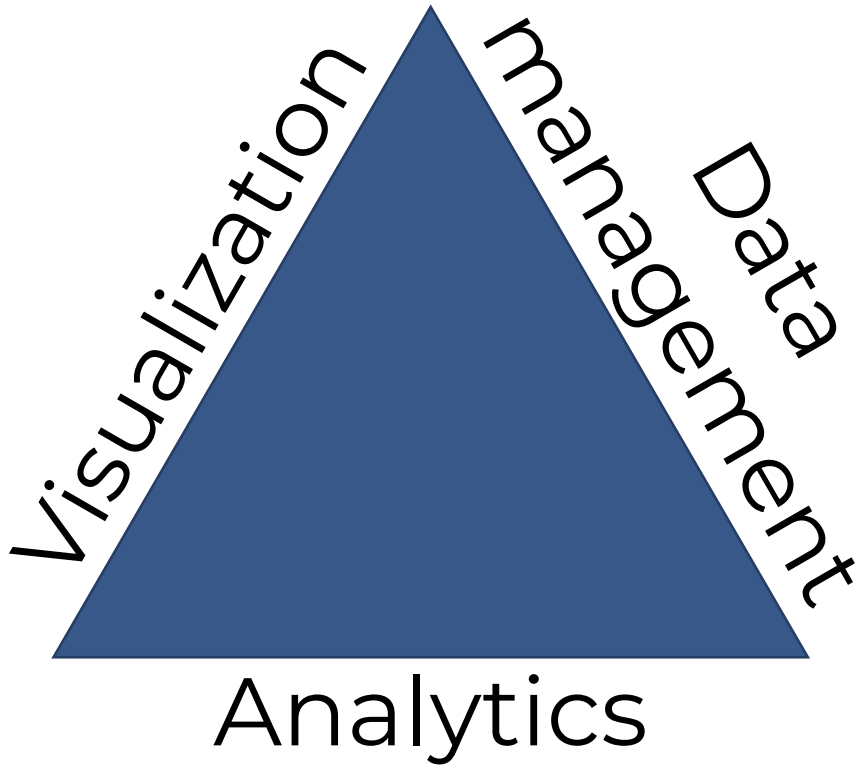
Fabio Miranda



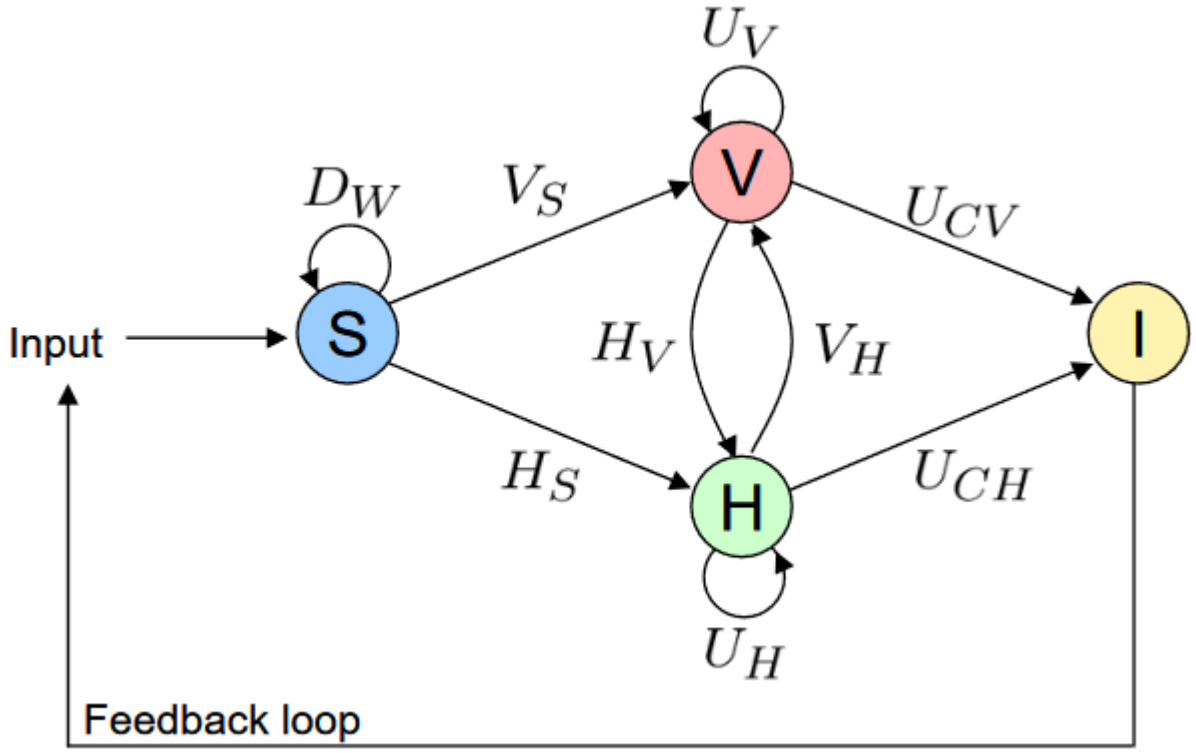




Visual analytics



Visual analytics process



["Visual Analytics: Scope and Challenges", Kleim et al., 2008]

$F: S \rightarrow I$
$f \in \{D_w, V_x, H_y, U_z\}$
S: data, I: insight

D_w: data pre-processing functionalities (data transformation, data cleaning, data selection, data integration)

$V_x, x \in \{S, H\}$: visualization functions
- V_S: functions visualizing data ($S \rightarrow V$)
- V_H: functions visualizing hypotheses ($\mathrm{H} \rightarrow V$)

$H_y, y \in \{S, V\}$: hypotheses generation process
- H_S: hypotheses from automatic process ($S \rightarrow H$)
- H_v: hypotheses from visualization ($\mathrm{V} \rightarrow H$)

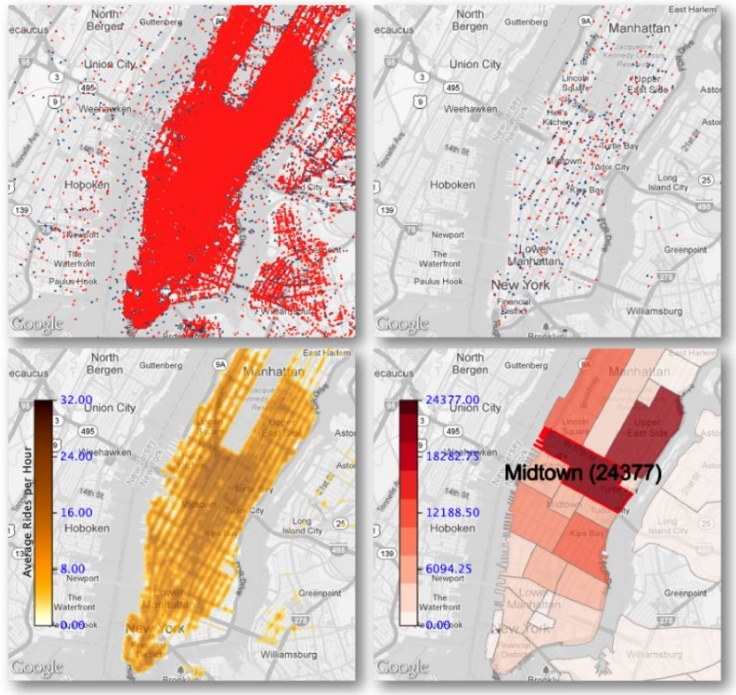
$U_z, z \in \{V, H, CV, CH\}$: user interactions
- U_V: visualization interaction ($V \rightarrow V$)
- U_H: hypotheses interaction ($\mathrm{H} \rightarrow H$)

U_{cv}: insight concluded from visualization

U_{ch}: insight concluded from hypothesis

Visual analysis process: taxi data





Distribution of NYC Taxi Pickups and Dropoffs in Manhattan

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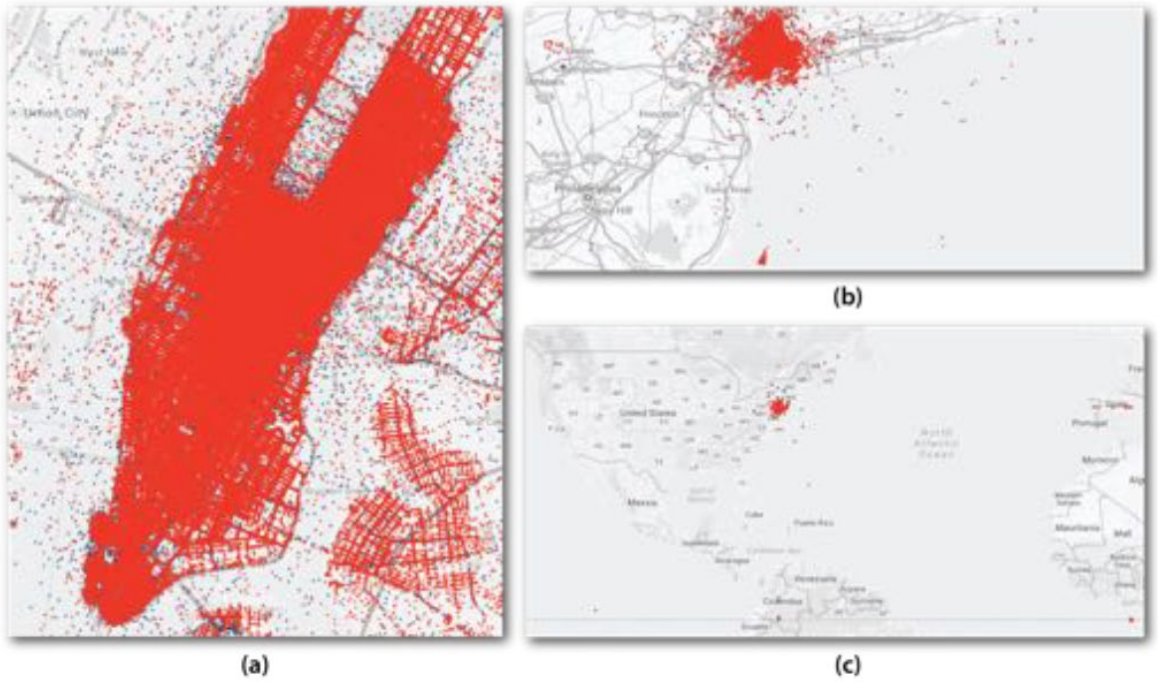
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Visual analysis process: taxi data

- Data: ~500,000 trips / day; ~1,000,000,000 trips in 5 years
  - Spatiotemporal: pickup + dropoff
  - Trip attributes: distance traveled, fare, tip, etc.
- Government, policy makers and scientists are usually unable to *interactive* explore the *whole* data.
  - Too many slices to examine.

Visual analysis process: taxi data

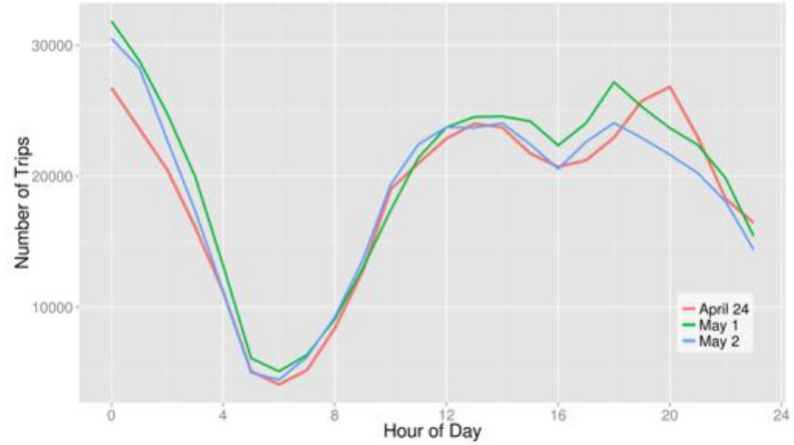


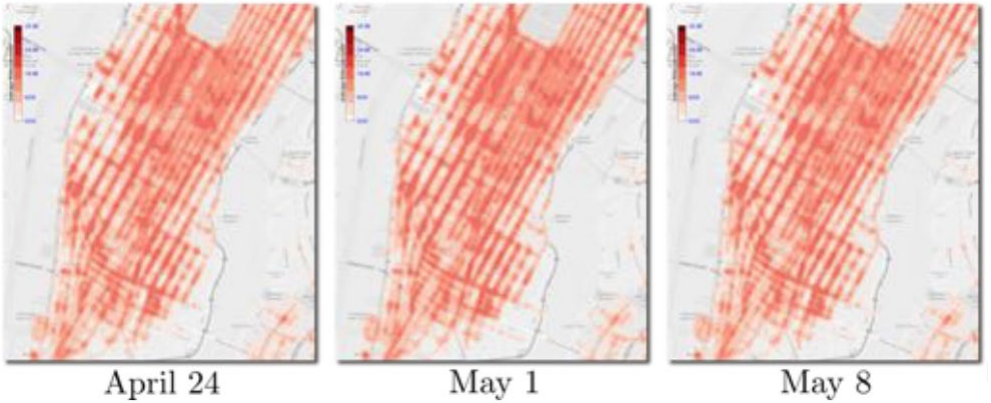
- Taxi pickups and drop-offs: data sets $S_1, S_2, \ldots, S_n$
- Pre-processing
  - Data cleaning (e.g., remove points outside NYC): $d_c(S_1, \ldots, S_n)$
  - Data integration (e.g., add weather column): $d_i(S_1, \ldots, S_n)$
  - Data transformation (e.g., raw csv's to binary): $d_t(S_1, \ldots, S_n)$
  - Data selection (e.g., only Manhattan trips): $d_s(S_1, \ldots, S_n)$

Visual analysis process: taxi data



- Too many slices
- 365*24 1-hour slices in just one year
- Which slices are interesting?

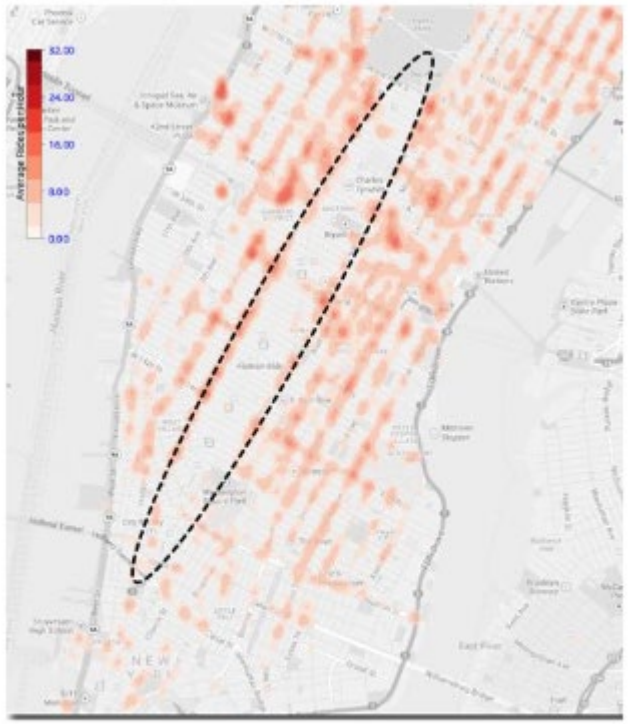




Visual analysis process: taxi data



- Too many slices
- 365*24 1-hour slices in just one year
- Which slices are interesting?



May 1 (8-9am)

Visual analysis process: taxi data



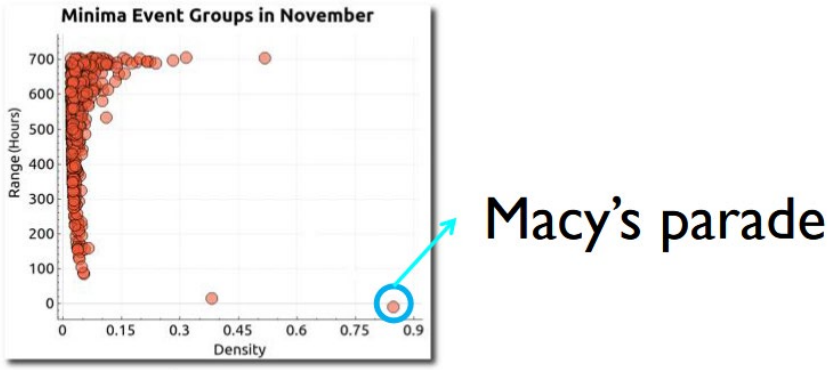
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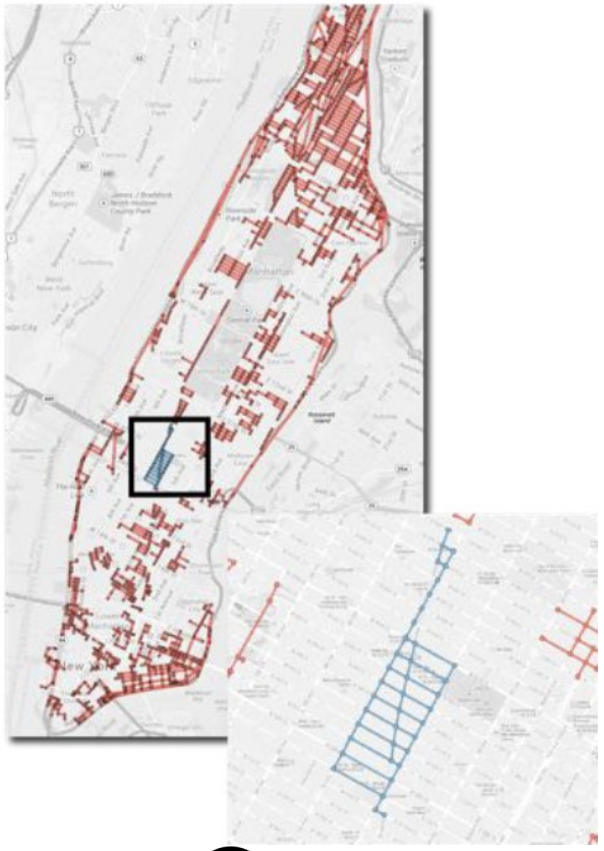
- Guide users towards *interesting* data slices
- Data mining:
  - Identifying events with arbitrary spatial structure at multiple temporal scales:

$$h' = h_s(S')$$
$$with\ S' = d_s(d_t(d_i(d_c(S_1, \ldots, S_n))))$$

Visual analysis process: taxi data

- Visualization:
  - Visualizing the previous hypothesis: $v' = v_h(h')$
  - Adjusting parameters: $v'' = u_v(v')$
  - Hypotheses from visualization: $h'' = h_v(v'')$, leading to an insight (unusual taxi activity around Macy's parade).





Visual analysis process: taxi data

- Taxi pickups and drop-offs: data sets $S_1, S_2, \ldots, S_n$
- Pre-processing
  - Data cleaning (e.g., removing): $d_c(S_1, \ldots, S_n)$
  - Data integration (e.g., adding extra attributes): $d_i(S_1, \ldots, S_n)$
  - Data transformation (e.g., format suitable for analysis): $d_t(S_1, \ldots, S_n)$
  - Data selection (e.g., subset of sensors): $d_s(S_1, \ldots, S_n)$
- Data mining:
  - Identifying events with arbitrary spatial structure at multiple temporal scales:

$$h' = h_s(S'), with\ S' = d_s\left(d_t\left(d_i(d_c(S_1, \ldots, S_n))\right)\right)$$

- Visualization:
  - Visualizing the previous hypothesis: $v' = v_h(h')$
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  - Hypotheses from visualization: $h'' = h_v(v'')$

$$h_v(u_v(v_h(h_s(d_s(d_t(d_i(d_c(S_1, \ldots, S_n)))))))))$$

Visual analytics mantra

Visualization mantra:

"*Overview first, zoom / filter, details on demand*"

[Shneiderman, 1996]

Visual analytics mantra:

"*Analyze first - show the important - zoom, filter and analyze further - details on demand*"

[Kleim et al., 2008]

Visual analytics + big data

"We propose a new model that allows users to visually query taxi trips. Besides standard analytics queries, the model supports origin-destination queries that enable the study of mobility across the city. We show that this model is able to express a wide range of spatiotemporal queries, and it is also flexible in that not only can queries be composed but also different aggregations and visual representations can be applied, allowing users to explore and compare results."



Visual Exploration of Big Spatio-Temporal Urban Data: A Study of New York City Taxi Trips

Nivan Ferreira, Jorge Poco, Huy T. Vo, Juliana Freire, and Cláudio T. Silva

Fig. 1. Comparison of taxi trips from Lower Manhattan to JFK and LGA airports in May 2011. The query on the left selects trips that occurred on Sundays, while the one on the right selects trips that occurred on Mondays. Users specify these queries by visually selecting regions on the map and connecting them. In addition to inspecting the results depicted on the map, i.e., the dots corresponding to pickups (blue) and dropoffs (orange) of the selected trips, they can also explore the results through other visual representations. The scatter plots below the maps show the relationship between hour of the day and trip duration. Points in the plots are colored according to the spatial constraint represented by the arrows between the regions: trips to JFK in blue, and trips to LGA in red. The plots show that many of the trips on Monday between 3PM and 5PM take much longer than trips on Sundays.

Abstract—As increasing volumes of urban data are captured and become available, new opportunities arise for data-driven analysis that can lead to improvements in the lives of citizens through evidence-based decision making and policies. In this paper, we focus on a particularly important urban data set: taxi trips. Taxis are valuable sensors and information associated with taxi trips can provide unprecedented insight into many different aspects of city life, from economic activity and human behavior to mobility patterns. But analyzing these data presents many challenges. The data are complex, containing geographical and temporal components in addition to multiple variables associated with each trip. Consequently, it is hard to specify exploratory queries and to perform comparative analyses (e.g., compare different regions over time). This problem is compounded due to the size of the data—there are on average 500,000 taxi trips each day in NYC. We propose a new model that allows users to visually query taxi trips. Besides standard analytics queries, the model supports origin-destination queries that enable the study of mobility across the city. We show that this model is able to express a wide range of spatio-temporal queries, and it is also flexible in that not only can queries be composed but also different aggregations and visual representations can be applied, allowing users to explore and compare results. We have built a scalable system that implements this model which supports interactive response times; makes use of an adaptive level-of-detail rendering strategy to generate clutter-free visualization for large results; and shows hidden details to the users in a summary through the use of overlay heat maps. We present a series of case studies motivated by traffic engineers and economists that show how our model and system enable domain experts to perform tasks that were previously unattainable for them.

Index Terms—Spatio-temporal queries; urban data; taxi movement data; visual exploration

1 Introduction

For the first time in history, more than half of the world's population lives in urban areas. Enabling cities to deliver services effectively, efficiently, and sustainably is among the most important undertakings in this century. While in the recent past, decision makers and social scientists faced significant constraints in obtaining the data needed to understand city dynamics and evaluate policies and practices, data are now abundant. Many cities have started to make a wide range of data sets available, see e.g., [24, 10, 7]. The challenge now is how to make sense of these data.

We examine one particularly important urban data set: taxi trips. In New York City, each day 13,000 taxis carry over one million passengers and make, on average, 500,000 trips—totaling over 170 million trips a year. Taxi trips are thus valuable sensors of city life. Consider the plot in Fig. 2, which shows how the number of trips per day varies

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Manuscript received 31 March 2013; accepted 1 August 2013; posted online 13 October 2013; mailed on 4 October 2013. For information on obtaining reprints of this article, please send e-mail to: tvcg@computer.org.

[Ferreira et al., 2013]

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TaxiVis

- Interaction capabilities that enable users to pose queries over all the dimensions of the data and flexibly explore the associated attributes.
- Compare spatiotemporal slices through multiple coordinated views.
- Interactively compose and refine queries, and generalize them by performing parameter sweeps.
- Efficient data stored, and adaptive level-of-detail rendering to provide clutter-free visualizations.



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[Ferreira et al., 2013]

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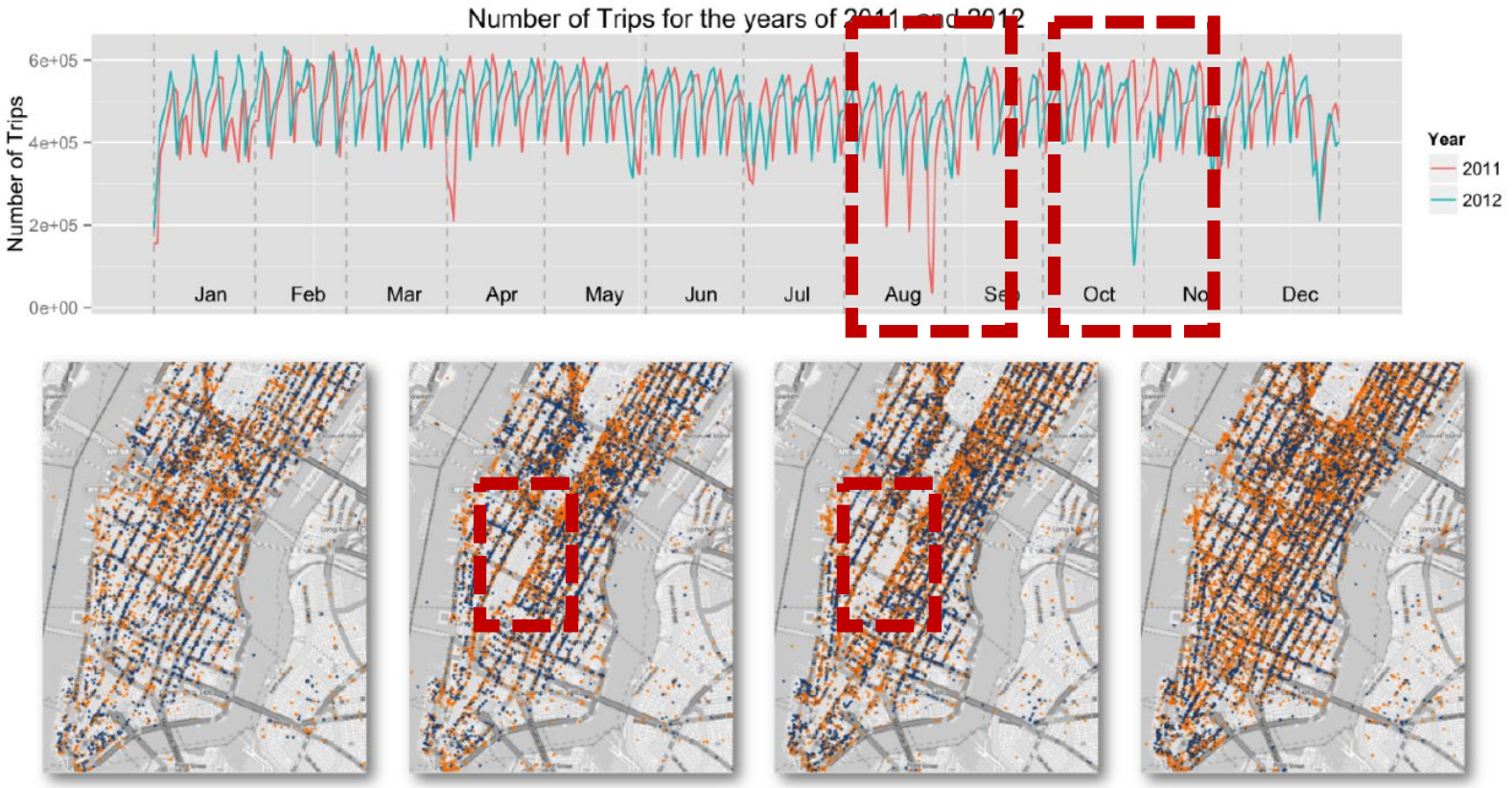
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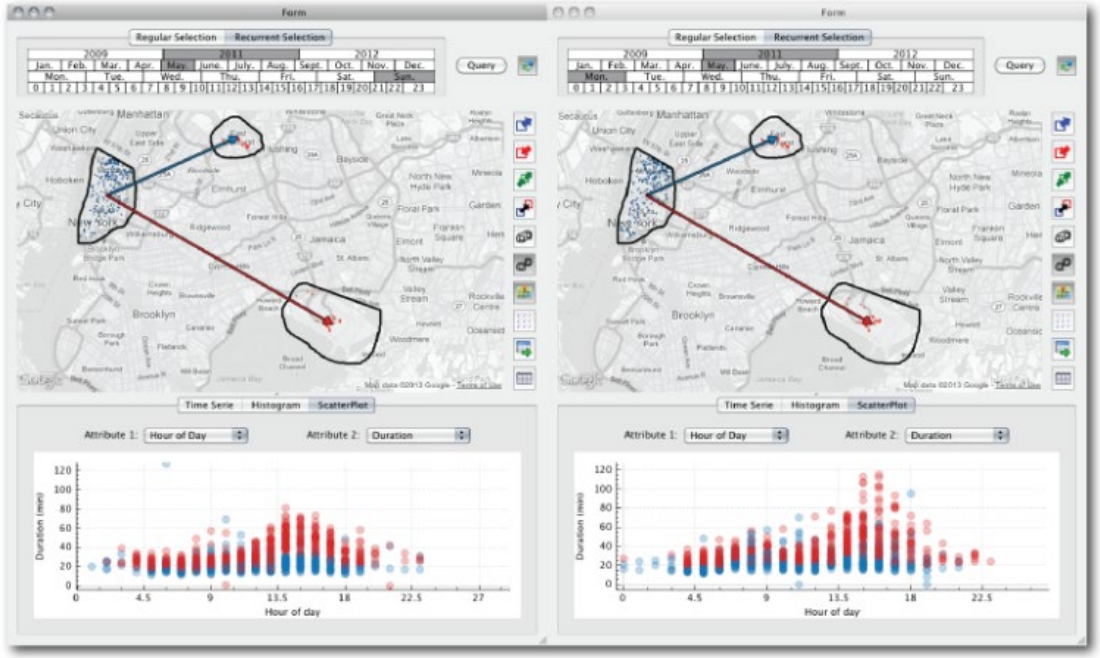
Taxi as sensors of city life

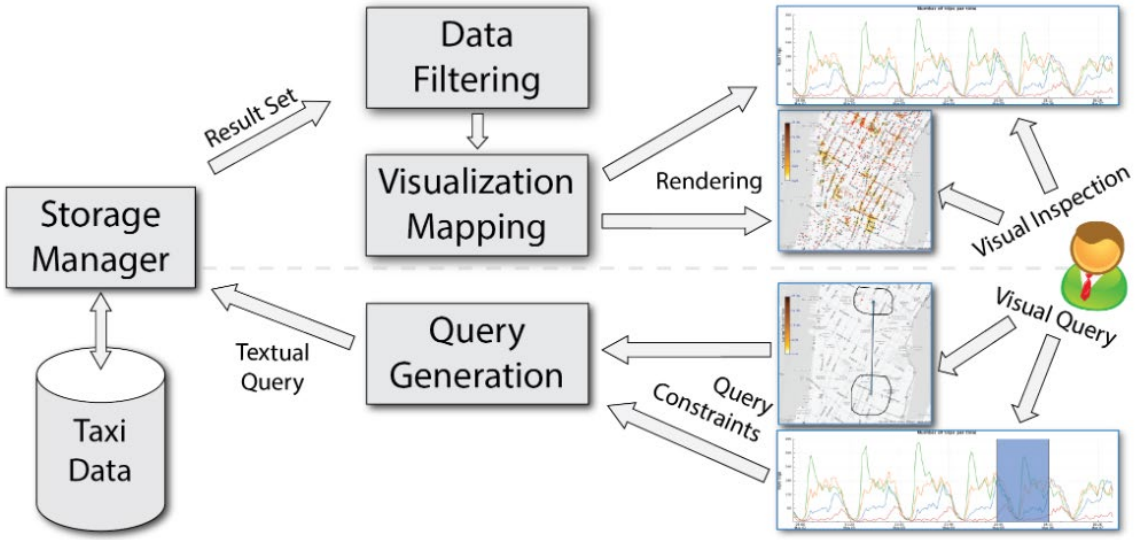
- Understanding the dynamics of the city, how different aspects of the data vary over space and time:
  - "*What is the average trip time from Midtown to the airports during weekdays?*"
  - "*How does the movement changes between Midtown and JFK throughout the day, over different days of the week?*"
  - "*How does the taxi fleet activity vary during weekdays?*"
- Ability to quickly test hypotheses: starting with one query about a specific place ("*movement patterns between Midtown and JFK*") and generalize to all neighborhoods.

Taxi as sensors of city life



TaxiVis



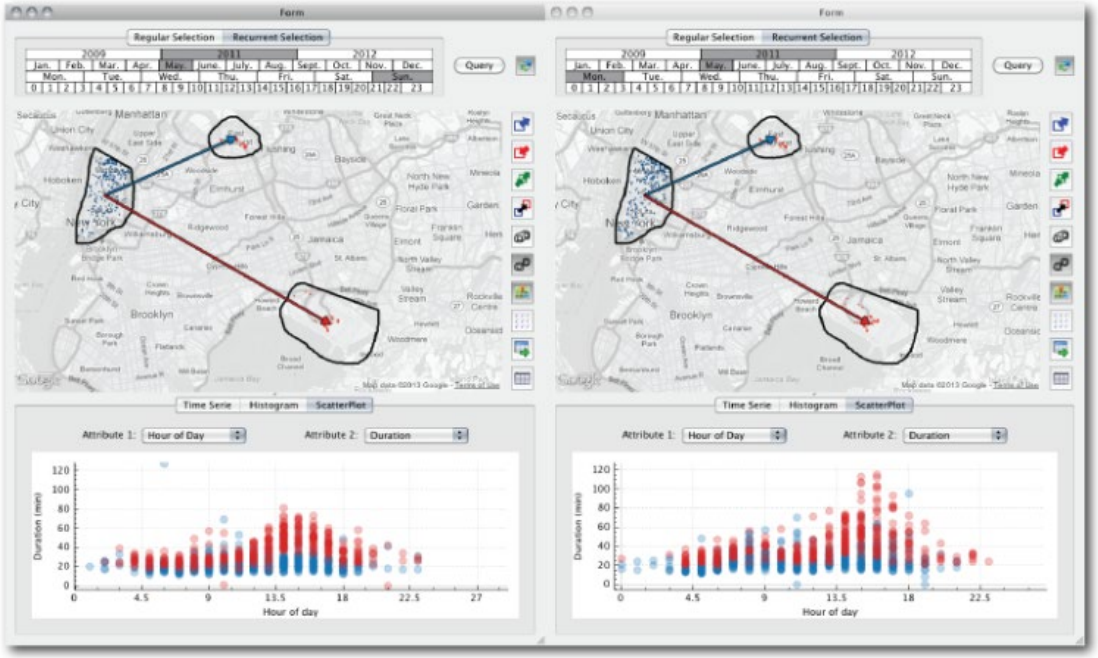


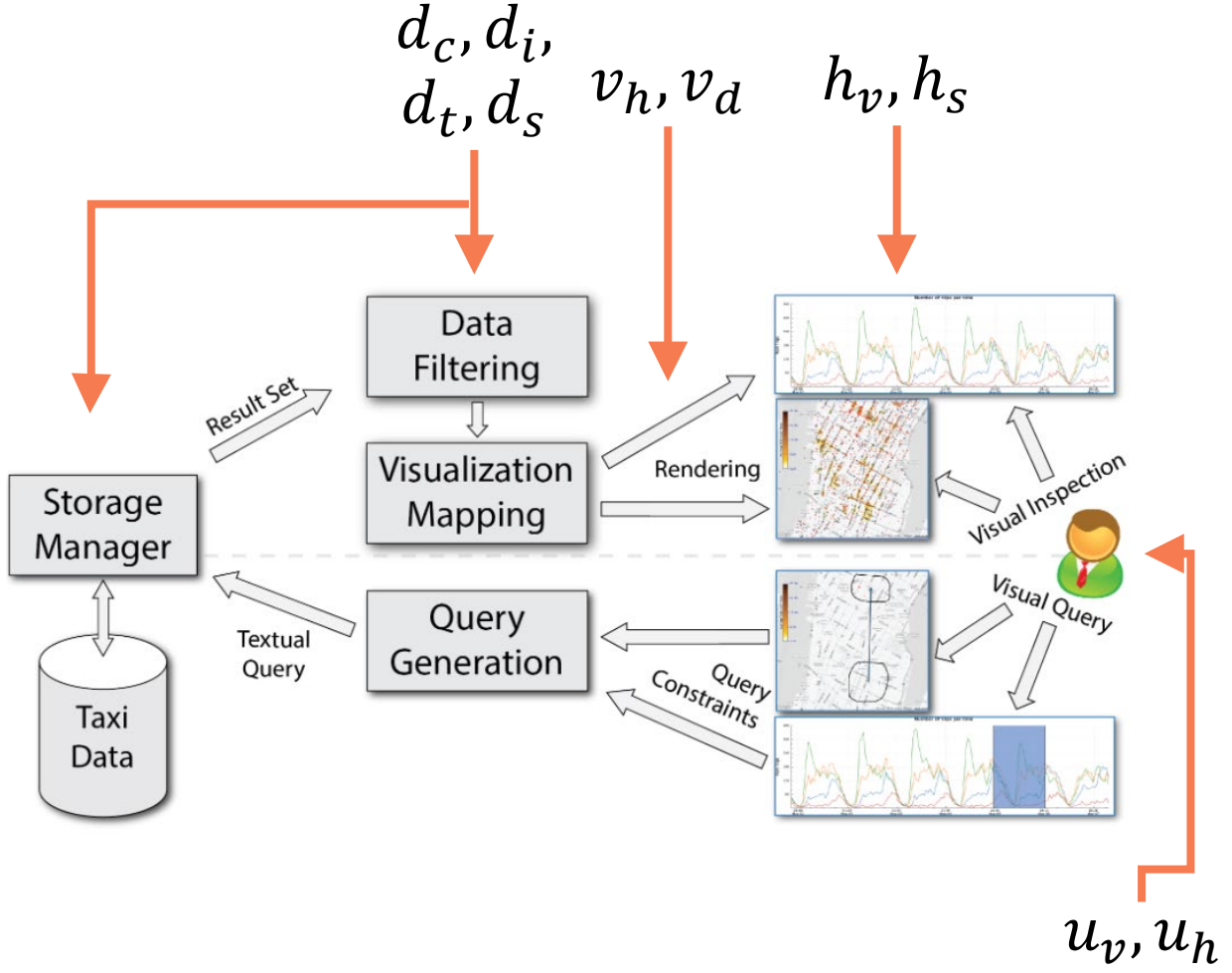
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TaxiVis


Form
Regular Selection
Recurrent Selection
Query
Time Serie
Histogram
ScatterPlot
Attribute 1:
Hour of Day
Attribute 2:
Duration
Duration (min)
Hour of day


$d_c, d_i,$
d_t, d_s
v_h, v_d
h_v, h_s
Data Filtering
Visualization Mapping
Query Generation
Storage Manager
Taxi Data
Result Set
Textual Query
Rendering
Visual Inspection
Visual Query
Query Constraints
u_v, u_h

Urban Mosaic: street-level images

“ … a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting.”



Urban Mosaic: Visual Exploration of Streetscapes Using Large-Scale Image Data

Fabio Miranda[1], Maryam Hosseini[2], Marcos Lage[3], Harish Doraiswamy[1], Graham Dove[1], Cláudio T. Silva[1]
[1]New York University; [2]Rutgers University; [3]Universidade Federal Fluminense
[1]{fmiranda,harishd,grahamdove,csilva}@nyu.edu; [2]mary.hosseini@rutgers.edu; [3]mlage@ic.uff.br

ABSTRACT
Urban planning is increasingly data driven, yet the challenge of designing with data at a city scale and remaining sensitive to the impact at a human scale is as important today as it was for Jane Jacobs. We address this challenge with Urban Mosaic, a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting. In doing so, we demonstrate how tools such as this might provide a bridge between the city and the street, by supporting activities such as visual comparison of geographically distant neighborhoods, and temporal analysis of unfolding urban development.

Author Keywords
Urban planning; Interactive visualization; Data analysis; Urban data

CCS Concepts
•**Human-centered computing** → **Human computer interaction (HCI);** *Visual analytics; Visualization toolkits;*

"A sense of place is built up, in the end, from many little things too, some so small people take them for granted, and yet the lack of them takes the flavor out of the city."

(Jane Jacobs, Downtown is for People)

INTRODUCTION
For those of us living in or visiting the world's major cities, their dynamism and complexity are immediately apparent. Yet urban planners and designers must work in a context where any single intervention, perhaps aimed at altering just one aspect, can have a wide ranging impact on a variety of interrelated components [12], affecting things at both a macro *city scale* and a micro *human scale*. Examples of changes at a city scale include, suburbanization, economic deconcentration, modification to transport infrastructure, rezoning and/or gentrification of neighborhoods, and major renewal projects. Examples of changes at a human scale, on the other hand, are reflected in the city's urban fabric, and include aspects that make a city livable, encourage walking, and contribute to the perception of safety; in other words affect the day to day lives of its inhabitants. This might be manifested in lighting, shadow, sky exposure, open-front shops, the details of building facades, etc. [32].

Responding to these challenges at the city scale, planners and designers have been aided by a rapid growth in data from urban environments, and so are able to turn to computational methods and large-scale data analysis, which increase understanding by quantifying different aspects of the city (e.g., [36, 13, 14, 29, 70, 67, 48]). However, while sensitivity to the impact of change at a human scale remains as important today as it was for Jane Jacobs and others in the 1950s and 1960s [47], analyses of suitable data, which emphasize qualitative, visual details, are often difficult and time-consuming to perform. Although there is an increasing availability of street-view images, which support a degree of virtual assessment and auditing of the built environment, their distribution is often temporally sparse and so analysis is limited.

This paper introduces Urban Mosaic, a tool for visually exploring the urban fabric. It responds to the challenges practitioners face by employing a newly available spatially and temporally dense data set of street-level images from New York City (NYC) Urban Mosaic is a visual exploration system designed to help practitioners in urban planning and design gain insight into the human scale impact of changes in the urban fabric. It utilizes state-of-the-art computer vision techniques for image similarity search and clustering, together with efficient spatio-temporal selection and aggregation over the image metadata to visually explore and map this image data set. It further allows the analysis to be augmented using spatio-temporal urban data from a variety of other sources (e.g., census, transport, crime, weather, housing market, zoning, noise complaints). The image data set used in this work contains 7.7 million images captured in the Manhattan and Brooklyn boroughs between April 2016 and April 2017 using car-mounted cameras. Urban Mosaic has been developed as a collaboration between researchers in urban planning, visual analytics, and HCI. We include a detailed

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Urban Mosaic: street-level images

- Demonstrates the potential for images from large-scale street-view data sets to help bridge data-driven urban planning and design at city scale and at the human level.
- Exploration of the urban fabric using large spatiotemporal collection of images.
- Visual comparison of geographically distant areas to analyze unfolding urban developments.



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For those of us living in or visiting the world's major cities, their dynamism and complexity are immediately apparent. Yet urban planners and designers must work in a context where any single intervention, perhaps aimed at altering just one aspect, can have a wide ranging impact on a variety of interrelated components [12], affecting things at both a macro *city scale* and a micro *human scale*. Examples of changes at a city scale include, suburbanization, economic deconcentration, modification to transport infrastructure, rezoning and/or gentrification of neighborhoods, and major renewal projects. Examples of changes at a human scale, on the other hand, are reflected in the city's urban fabric, and include aspects that make a city livable, encourage walking, and contribute to the perception of safety; in other words affect the day to day lives of its inhabitants. This might be manifested in lighting, shadow, sky exposure, open-front shops, the details of building facades, etc. [32].

Responding to these challenges at the city scale, planners and designers have been aided by a rapid growth in data from urban environments, and so are able to turn to computational methods and large-scale data analysis, which increase understanding by quantifying different aspects of the city (e.g., [36, 13, 14, 29, 70, 67, 48]). However, while sensitivity to the impact of change at a human scale remains as important today as it was for Jane Jacobs and others in the 1950s and 1960s [47], analyses of suitable data, which emphasize qualitative, visual details, are often difficult and time-consuming to perform. Although there is an increasing availability of street-view images, which support a degree of virtual assessment and auditing of the built environment, their distribution is often temporally sparse and so analysis is limited.

This paper introduces Urban Mosaic, a tool for visually exploring the urban fabric. It responds to the challenges practitioners face by employing a newly available spatially and temporally dense data set of street-level images from New York City (NYC) Urban Mosaic is a visual exploration system designed to help practitioners in urban planning and design gain insight into the human scale impact of changes in the urban fabric. It utilizes state-of-the-art computer vision techniques for image similarity search and clustering, together with efficient spatio-temporal selection and aggregation over the image metadata to visually explore and map this image data set. It further allows the analysis to be augmented using spatio-temporal urban data from a variety of other sources (e.g., census, transport, crime, weather, housing market, zoning, noise complaints). The image data set used in this work contains 7.7 million images captured in the Manhattan and Brooklyn boroughs between April 2016 and April 2017 using car-mounted cameras. Urban Mosaic has been developed as a collaboration between researchers in urban planning, visual analytics, and HCI. We include a detailed

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[Miranda et al., 2020]

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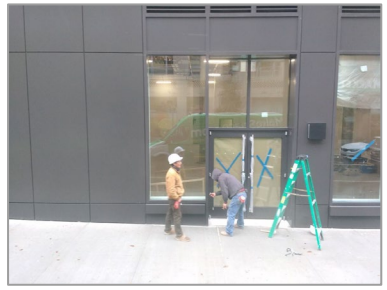
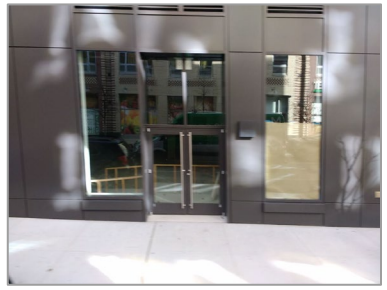
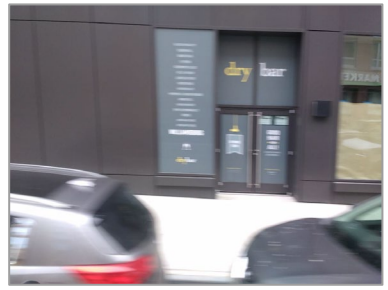
Temporally-dense street-level images









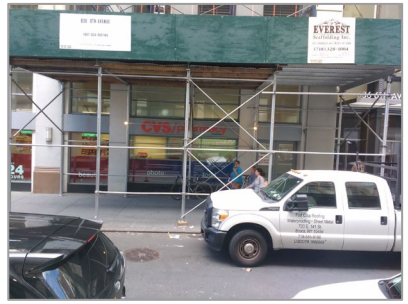
Objectives

Support the interactive analysis of the city at the micro scale, over geographically distant regions.





Comparison of the urban fabric in different regions





Assessment of features in the built environment

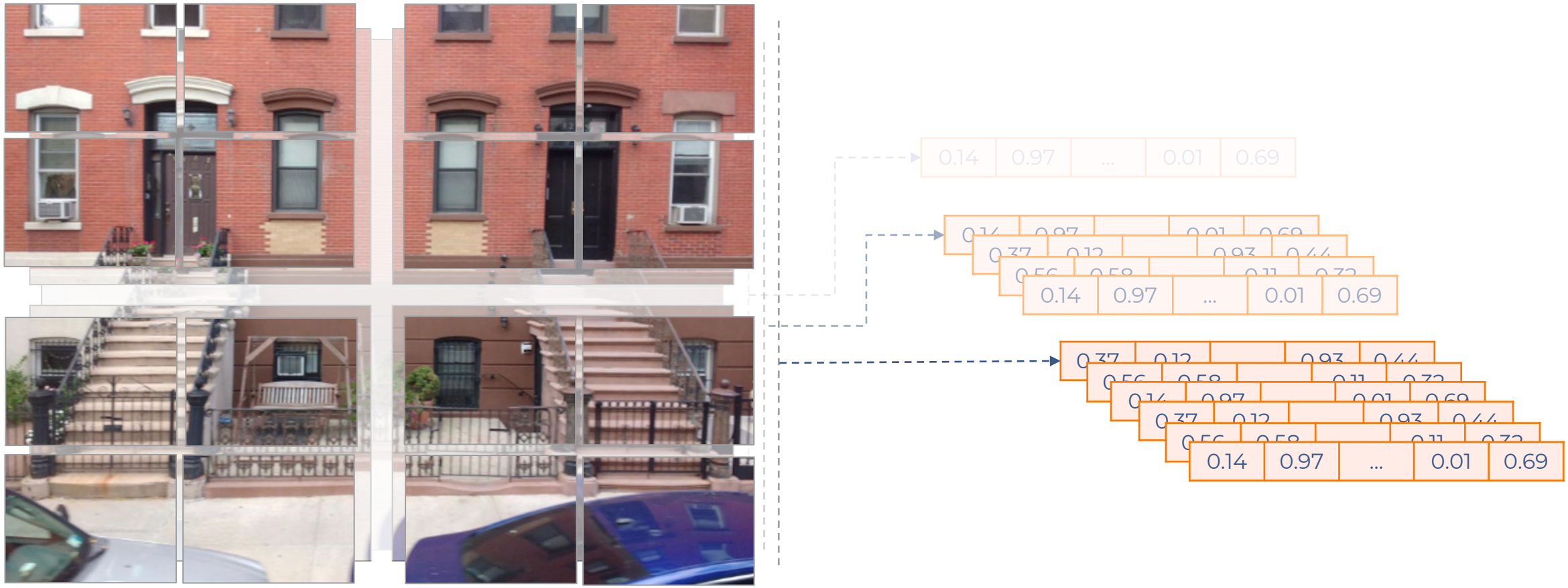


Assessment of walkability and accessibility

Image query composition

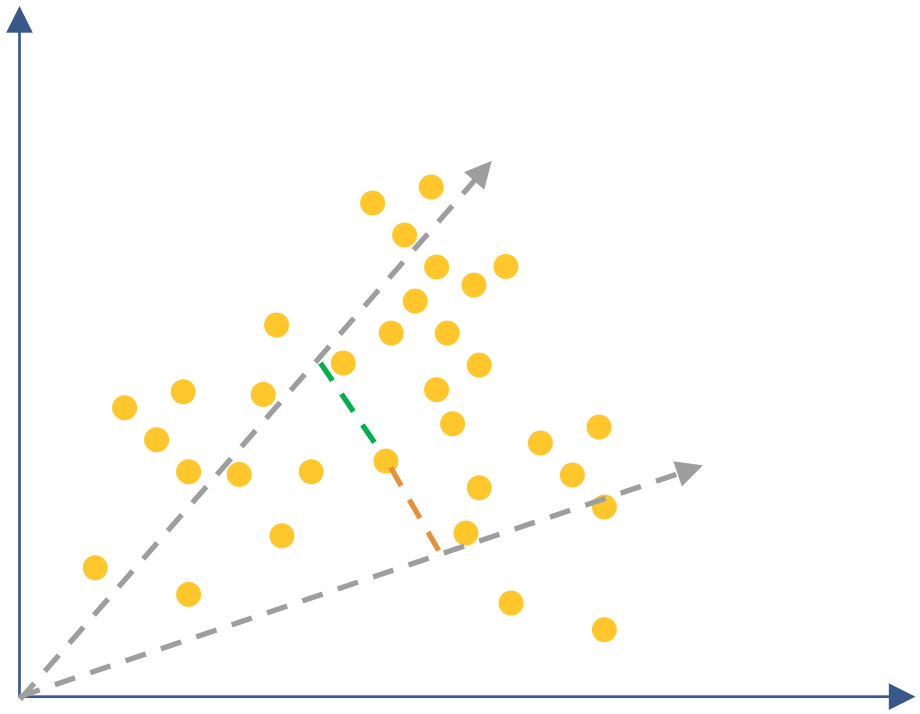


Image embeddings



- Locality sensitive hashing

  21 GB worth of embeddings data



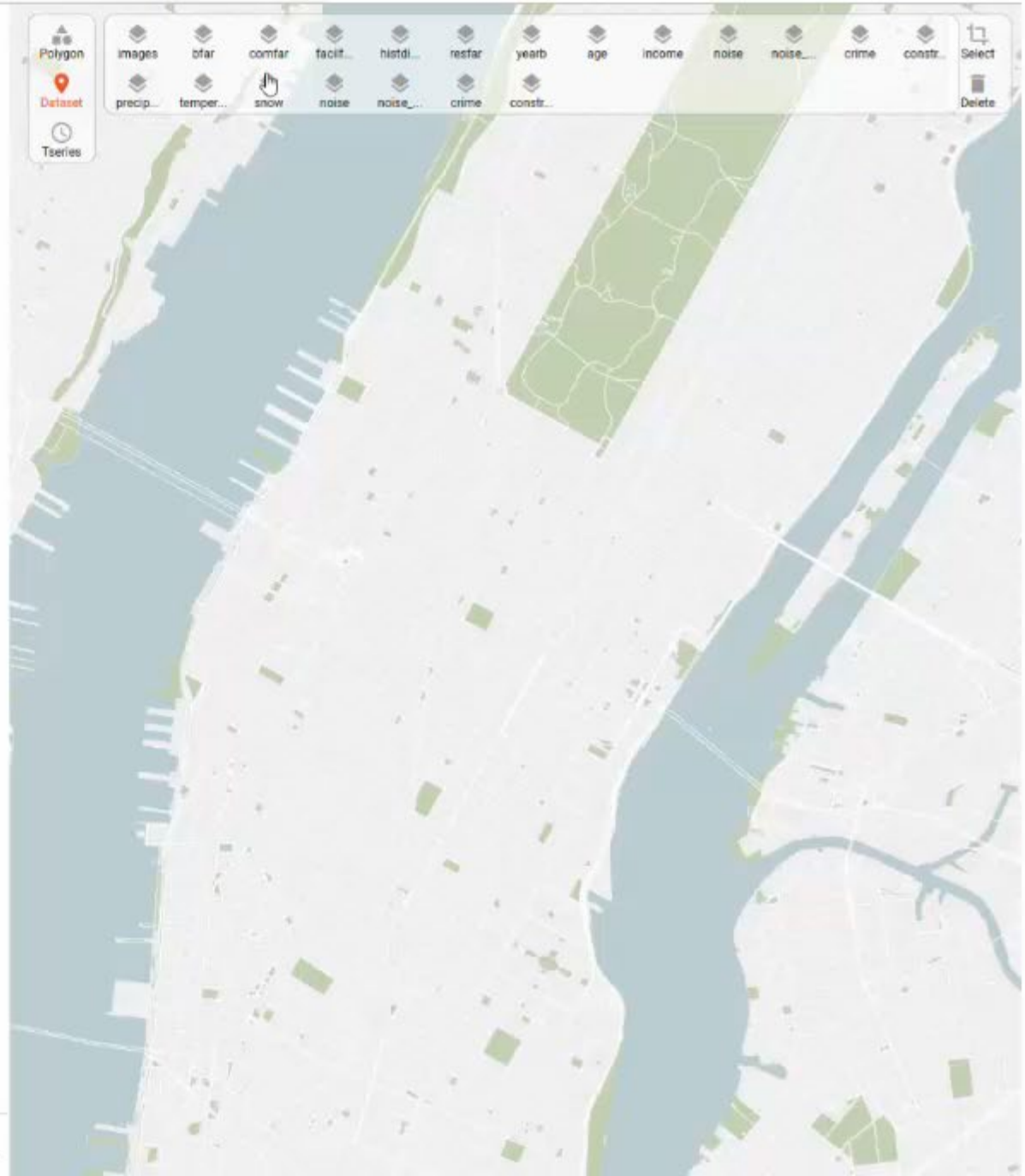
0.37	0.12	...	0.93	0.44

4 kbytes per embedding

0	0	...	1	0

64 bytes per embedding

$$\alpha_{1,2} = \cos^{-1}\left(\frac{\overrightarrow{v_1} \cdot \overrightarrow{v_2}}{|\overrightarrow{v_1}||\overrightarrow{v_2}|}\right)$$


Polygon
Dataset
Tseries
images
bfar
comfar
facil...
histdi...
resfar
yearb
age
income
noise
noise_...
crime
constr...
Select
Delete
precip...
temper...
snow
noise
noise_...
crime
constr...
time
ascending
time
ascending

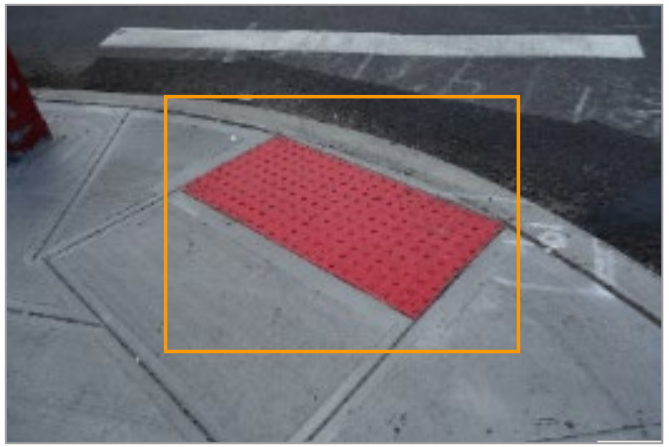
Accessibility for older adults



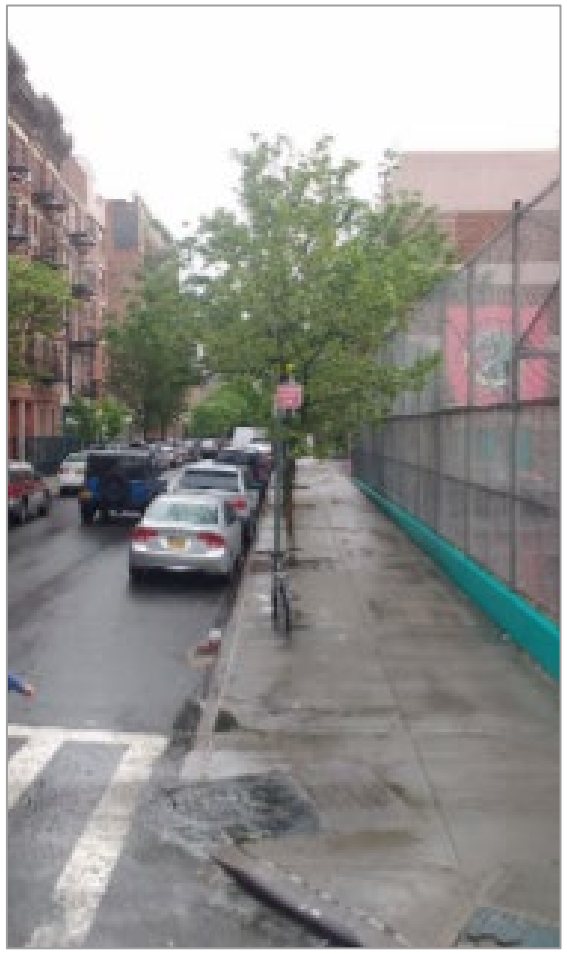
Focuses on interventions that enable older adults to “Age in Place” and prevent outdoor fall.

Impact of the neighborhood environment on falls.

Installation of tactile pavings



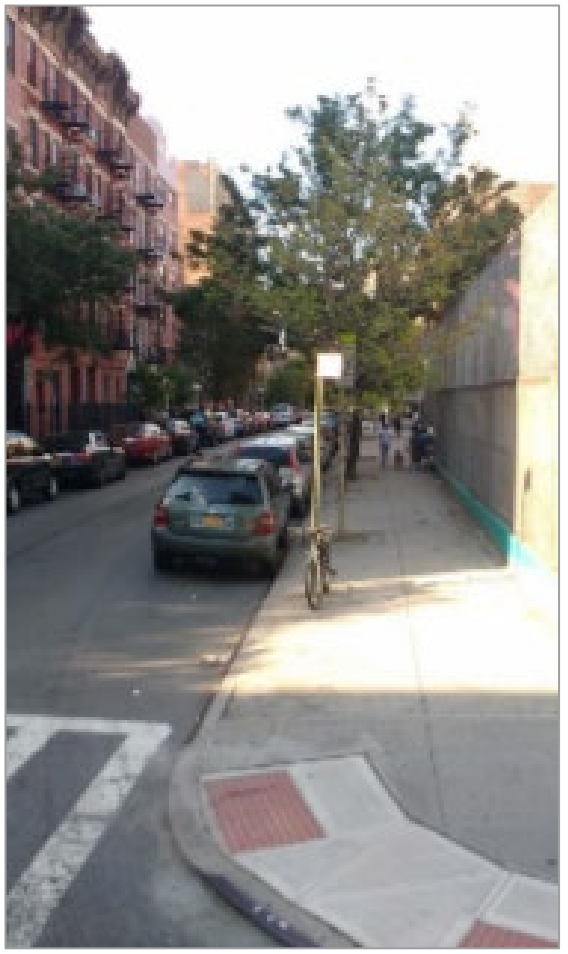




June

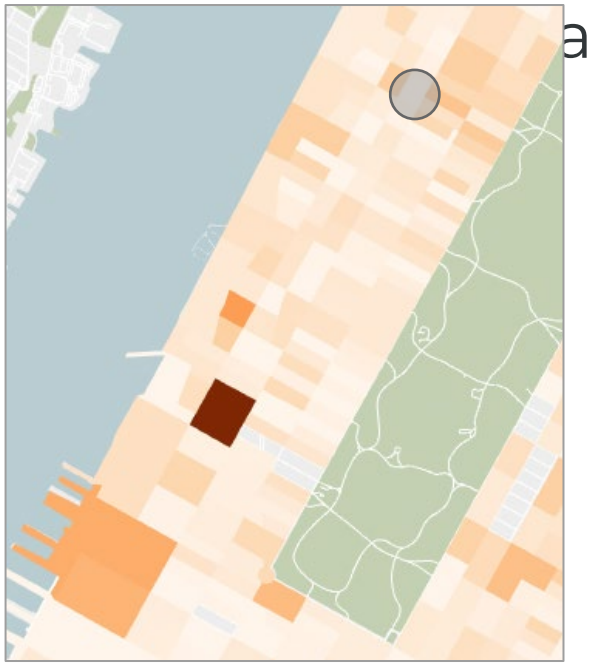


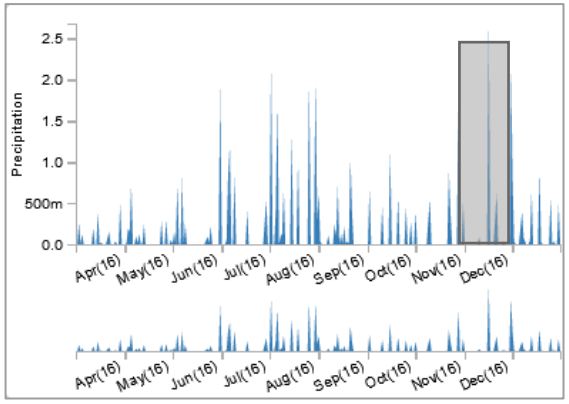
August

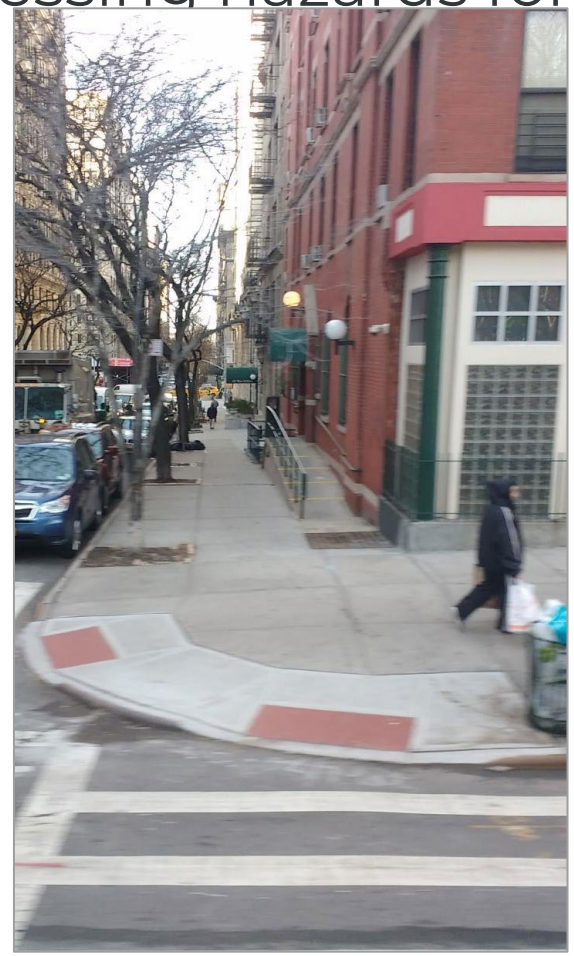


October

assessing hazards for older adults











Main takeaways

- General approaches might not scale.
- Data visual analytics might require structures specifically designed for a given task.
- Understand the data, and how it can be transformed to best fit different computational resources and architectures.

Visual analytics process: Urban Mosaic (data)

- Input: georeferenced, timestamped street-level images and urban data
- Data processing:
  - Feature computation: CNN descriptors for images and image regions (d_f)
  - Indexing: compressed feature representation for interactive search (d_i)

$$d_i(d_f(S_1, \ldots, S_n))$$

Visual analytics process: Urban Mosaic (analytics)

- Analytics:
  - Image similarity search and clustering (h_s)

$$h_s(d_i(d_f(S_1, \ldots, S_n)))$$

Visual analytics process: Urban Mosaic (visualization)

- Urban mosaic: clusters of visually similar street-level images
- Urban data interface: spatial heatmaps, selected clusters, time series.
- Interactive selection and visual comparison across locations and time.

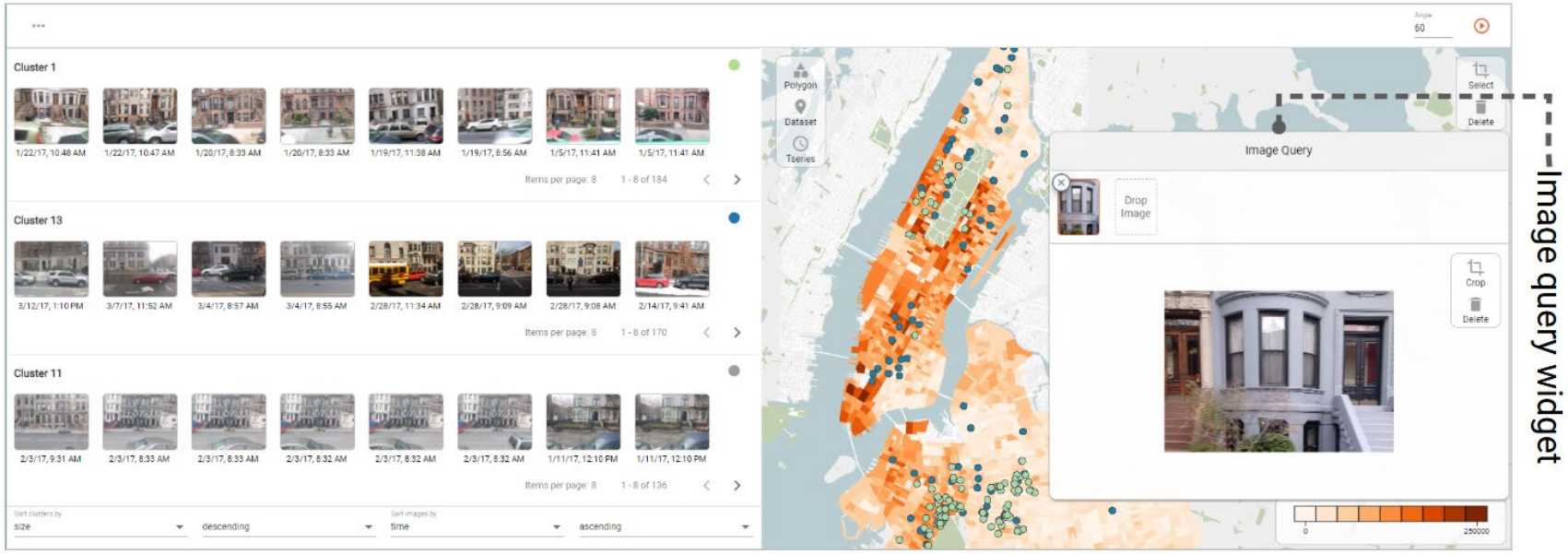


Image mosaic

Urban data interface

v_s : visualize spatial data
v_h: visualize candidate images and clusters
u_v : select, filter, and inspect
h_v: form or revise a hypothesis

Visual analytics process: Urban Mosaic (overall)

- Street-level images and urban attributes: $S_1, \ldots, S_n$
- Preprocess images and metadata; compute and index visual features: d_i, d_f
- Generate candidate patterns through search, clustering, and filtering: h_s
- Visualize results, inspect images, and refine queries iteratively: v_s, v_h, h_v, u_v

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$$S' = d_i\left(d_f(S_1, \ldots, S_n)\right)$$

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$$h' = h_s(S')$$

$$v' = v_h(h') \qquad v'' = u_v(v') \qquad h'' = h_v(v'')$$

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$$S' = d_i\left(d_f(S_1, \dots, S_n)\right)$$

$$h' = h_s(S')$$

$$v' = v_h(h') \qquad v'' = u_v(v') \qquad h'' = h_v(v'')$$

$$h'' = h_v(u_v(v_h(h_s(d_i(d_f(S_1, \dots, S_n))))))$$

Urban Mosaic: System architecture

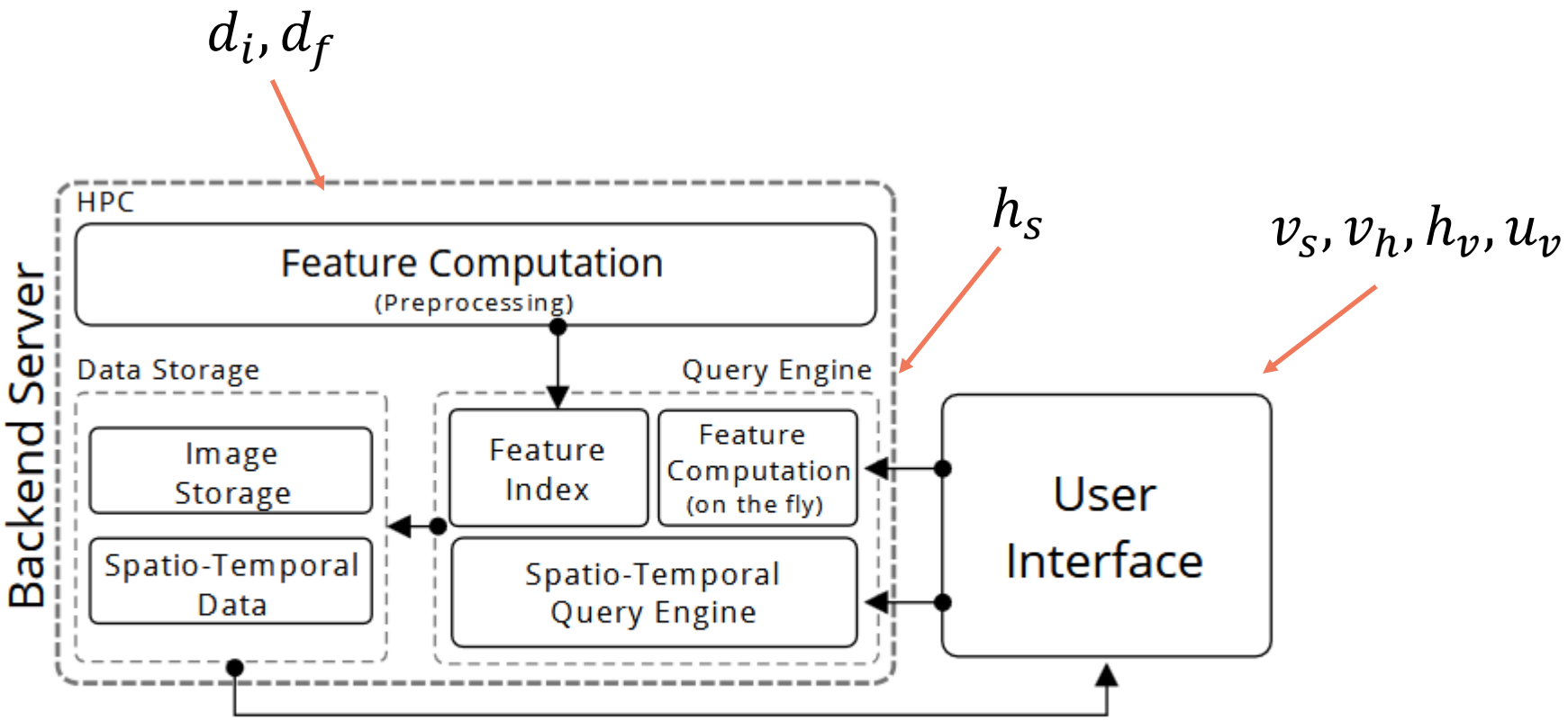


Figure 1. System architecture of the Urban Mosaic system.